

JEE MAIN REVISION MASTER: SALT ANALYSIS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 43 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Salt Analysis — Applied Cases, Reactions & Traps	26 PYQs	Applied Topics
Very High	Salt Analysis — Standard Calculations & Problem Archetypes	17 PYQs	Quantitative Analysis
Moderate	Salt Analysis — Core Principles & Governing Laws	6 PYQs	Core Principle

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Salt Analysis — Applied Cases, Reactions & Traps	OH^- , NH_4OH , Pb^{2+}
Salt Analysis — Standard Calculations & Problem Archetypes	OH^- , H_2S , Fe^{3+} , Pb^{2+} , Cu^{2+}
Salt Analysis — Core Principles & Governing Laws	Sr^{2+} , Ba^{2+} , Ca^{2+} , Cu^{2+} , Zn^{2+} , Co^{2+} , Fe^{2+} , NO_2^- , Sr^{2+} , Ba^{2+} , Ca^{2+} , Cu^{2+}

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Salt Analysis — Applied Cases, Reactions & Traps (Very High Yield — 26 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Salt Analysis.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2026 Morning Shift] Q#2 (MCQ): "A salt with few drops of conc. HCl gives apple green colour in flame test. The group precipitate of the salt is dissolved in acetic acid and..." [Ans: N/A]
- [2025 Evening Shift] Q#1 (MCQ): "Identify the inorganic sulphides that are yellow in colour : (A) $(\text{NH}_4)_2\text{S}$ (B) PbS (C) CuS (D) As_2S_3 (E) As_2S_5 Cho..." [Ans: N/A]
- [2025 Evening Shift] Q#2 (MCQ): "Match the LIST-I with LIST-II LIST-I (Reagent) LIST-II (Functional Group detected) A. Sodium bicarbonate solution I. double bond/unsaturatio..." [Ans: N/A]

3.2 Salt Analysis — Standard Calculations & Problem Archetypes (Very High Yield — 17 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Salt Analysis.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Among Fe^{3+} , Pb^{2+} , Cu^{2+} and Mn^{2+} , identify the one that gets precipitated out while passing H_2S in presence of NH_4OH ..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "Consider three metal chlorides x, y and z, where x is water soluble at room temperature, y is sparingly soluble in water at room temperatu..." [Ans: N/A]
- [2026 Evening Shift] Q#4 (MCQ): "In the Group analysis of cations, Ba^{2+} & Ca^{2+} are precipitated respectively as..." [Ans: D]

3.3 Salt Analysis — Core Principles & Governing Laws (Moderate Yield — 6 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Salt Analysis.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Salt Analysis.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Salt Analysis PYQs.

Representative Source PYQs:

- [2026 Morning Shift] Q#3 (MCQ): "Given below are two statements: Statement I : Griss-Ilosvay test is used for the detection of nitrite ion, which involves the use of sulphan..." [Ans: D]
- [2025 Evening Shift] Q#3 (MCQ): "Formation of $\text{Na}_4[\text{Fe}(\text{CN})_5\text{NOS}]$, a purple coloured complex formed by addition of sodium nitroprusside in sodium carbonate extract of s..." [Ans: N/A]
- [2025 Morning Shift] Q#4 (MCQ): "Given below are two statements : Statement I : Mohr's salt is composed of only three types of ions-ferrous, ammonium and sulfate. Statement ..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 4th April Evening Shift (Q#1)

Among Fe^{3+} , Pb^{2+} , Cu^{2+} and Mn^{2+} , identify the one that gets precipitated out while passing H_2S in presence of NH_4OH as group reagent. The highest possible oxidation state of the corresponding metal is

- (A) +3
- (B) +4
- (C) +2
- (D) +7

Correct Answer: None

Solution Analysis: While passing H_2S in the presence of NH_4OH , the medium becomes basic. In basic medium, the concentration of S^{2-} ions increases, so sulphides of some metal ions get precipitated. Now let us check the given ions: Fe^{3+} : does not precipitate here as sulphide in this group Pb^{2+} : belongs to earlier group and is precipitated as chloride, not here Cu^{2+} : precipitates as CuS Mn^{2+} : can precipitate as MnS in ammoniacal medium But in classical qualitative analysis, passing H_2S in presence of NH_4OH is used for precipitating group IV cations like Mn^{2+}

Benchmark Problem 2: JEE Main 2026 (Online) 28th January Morning Shift (Q#3)

Given below are two statements: Statement I : Griss-Ilosvay test is used for the detection of nitrite ion, which involves the use of sulphanilic acid and α -naphthylamine reagent. Statement II : In the above test, sulphanilic acid is diazotized by the acidified nitrite ion, which on further coupling with α -naphthylamine forms an azo-dye. In the light of the above statements, choose the correct answer from the options given below

- (A) Statement I is true but Statement II is false
- (B) Both Statement I and Statement II are false
- (C) Statement I is false but Statement II is true
- (D) Both Statement I and Statement II are true **[Correct]**

Correct Answer: D

Solution Analysis: In Gries-Ilosvay test for nitrite ion (NO_2^-), the reagents used are sulphanilic acid and α -naphthylamine. In acidic medium, nitrite ion forms nitrous acid HNO_2 , which diazotizes sulphanilic acid to give a diazonium salt. This diazonium salt then couples with α -naphthylamine to form a coloured azo dye (usually reddish/pink). So, Statement I is true and Statement II is also true. Correct option: D (Both Statement I and Statement II are true)

Benchmark Problem 3: JEE Main 2025 (Online) 28th January Evening Shift (Q#1)

Identify the inorganic sulphides that are yellow in colour : (A) $(\text{NH}_4)_2\text{S}$ (B) PbS (C) CuS (D) As_2S_3 (E) As_2S_5 Choose the correct answer from the options given below :

- (A) (D) and (E) only
- (B) (A) and (B) only
- (C) (A), (D) and (E) only
- (D) (A) and (C) only

Correct Answer: None

Solution Analysis: We want to identify which among the listed sulfides are yellow in color: $(\text{NH}_4)_2\text{S}$ Ammonium sulfide is typically encountered as an aqueous solution that can be colorless to slightly yellowish in solution form. It is not known as a distinct "yellow sulfide" compound in the same sense as solid colored sulfides like orpiment (see below). PbS (lead(II) sulfide) PbS (Galena) is well-known to be black or grayish black in color. CuS (copper(II) sulfide) CuS is black (covellite). As_2S_3 (arsenic(III) sulfide) As_2S_3 (orpiment) is characteristic